1 July 2005

ISA Certified Control Systems Technician (CCST) Program

This question is from the Level III study guide, Domain 2, Loop checking.

Level III represents a professional who has a 13-year total of education, training, or experience.

CCST question

Programmable Logic Controllers:

A. Use sequential logic control.

B. Perform PID control with the same capability as DCS.

C. Have limitations in flexibility of control strategies.

D. Are programmed using flow charts and P & IDs.

CCST answer

We use ladder logic to program programmable logic controllers (PLCs), where a series of complex logic checks are required before something is turned on. Ladder logic is useful for simple but critical control systems or for reworking old hardwired relay circuits. As PLCs became more sophisticated, it has also been used in very complex automation systems.

Ladder logic (sequential logic) is a rule-based language rather than a procedural language. A "rung" in the ladder represents a rule. When implemented with relays and other electromechanical devices, the various rules "execute" simultaneously and immediately.

The rules are typically executed sequentially by software in a loop. The best answer is **A**.

1 July 2005

ISA Certified Automation Professional (CAP) program

This question is from the CAP study guide, Performance Domain III, System Design.

Definition: Design, specify, and procure the hardware/software used in the system.

CAP question

Which of the following protection techniques is acceptable for equipment located in a Class I, Division 1 area?

A. Explosion-proof apparatus and non-incendive equipment

B. Explosion-proof apparatus and intrinsic safety

C. Dust ignition-proof and non-incendive equipment

D. Hermetically sealed and intrinsic safety

CAP answer

The best answer is **B**, explosion-proof apparatus and intrinsic safety.

Hazardous location protection techniques are defined by NEC 500.7 (NEC 2002 Code). The protection techniques are permitted as follows:

- Explosion proof apparatus—Class I, Division 1 or 2
- Intrinsic safety—Class I, Division 1 or 2; Class II, Division 1 or 2; or Class III, Division 1 or 2
- Nonincendive equipment—Class I, Division 2; Class II, Division 2; or Class III, Division 1 or 2 (does not include Class 1, Division 1)
- Dust ignition proof—Class II, Division 1 or 2 (does not include Class I, Division 1)
- Hermetically sealed—Class I, Division 2; Class II, Division 2; Class III, Division 1 or 2 (does not include Class I, Division 1)

1 August 2005

ISA Certified Control Systems Technician (CCST) Program

This question is from the Level III study guide, Domain 2, Loop checking.

Level III represents a professional who has a 13-year total of education, training, and/or experience.

CCST question

All of the following are basic components of a PLC system EXCEPT:

A. Processor

B. I/O system

C. Power supply

D. CRT monitor

CCST answer

A programmable logic controller (PLC) has a microprocessor, and it controls industrial processes. It communicates with other process control components through data links, handles simple switching tasks, PID (proportional-integral-derivative) control, complex data manipulation, arithmetic operations, timing, process, and machine control—all without a CRT (cathode ray tube) or video monitor. Engineers program it and leave it alone. The correct answer is D.

1 August 2005

ISA Certified Automation Professional (CAP) Program

This question is from the CAP study guide, Performance Domain III, System Design.

Definition: Design, specify, and procure the hardware/software used in the system.

CAP question

Which of the following control-valve body styles is MOST likely to develop blockage in a paper-pulp slurry service?

A. Pinch

B. Characterized ball

C. Butterfly

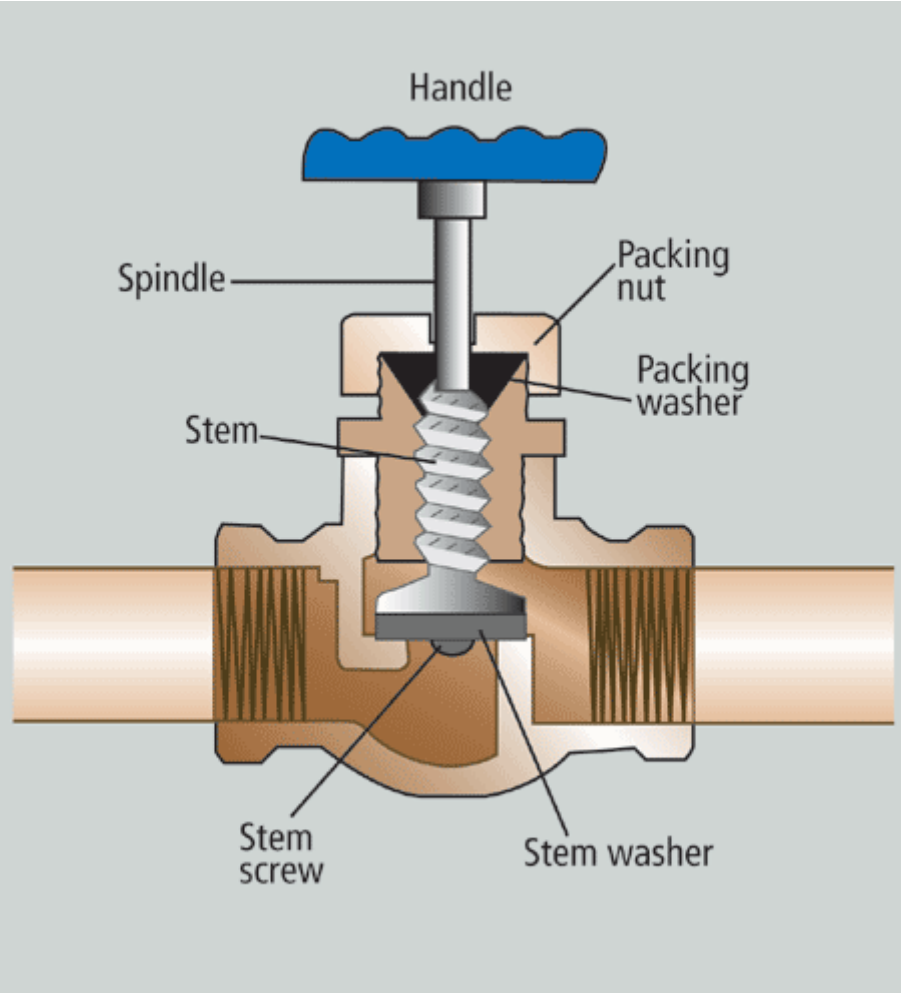
D. Globe

CAP answer

The correct answer is **D**, Globe.

A globe valve consists of a plug on the end of a stem that moves the plug onto a seat. Because of the body design of this valve, cavities in the body can easily fill with materials and develop blockage. Ball-, butterfly-, and pinch valve body style have straight-through flow, which is less likely to develop blockage.

References: Borden, Guy, *Control Valves: Practical Guides for Measurement and Control*, ISA Press, 1998. Lipták, Bela, *Instrument Engineers' Handbook: Process Control*, Volume 2. CRC Press, 2001.



1 September 2005

ISA Certified Control Systems Technician (CCST) Program

This question is from the Level III study guide, Domain 2, Loop checking.

Level III represents a professional who has a 13-year total of education, training, and/or experience.

CCST question

With a DCS it is possible to change from single loop control of a process variable to cascade, ratio, feedforward, adaptive, or multi-variable control by _____ if all of the peripheral hardware is installed in the field and connected to the DCS terminal board.

A. relocating the transmitters and FCE

B. reconfiguring the loop

C. installing a UPS and inverter

D. installing another CPU

CCST answer

In that all the peripherals and hardware are present and in order, the task of changing control strategies and algorithms and how the gathered data from the process undergoes manipulation by the distributed control system.

In ratio control, the DCS maintains a predetermined ratio between two variables. In cascade control, one controller output is the input for another controller. Feedforward input comes from some possible disturbance that then converts to corrective action to the process. Feedforward takes place outside the feedback loop. Adaptive and multivariable control strategies also leverage numbers per their own dispositions.

These control strategies use process data, process numbers, and manipulate those inputs mathematically. How that data parses depends on the configuration of the loop. The DCS doesn't care; it can do the arithmetic for all these control strategies.

The best answer is **B**.

1 September 2005

ISA Certified Automation Professional (CAP) Program

This question is from the CAP study guide, Performance Domain III, System Design.

Definition: Design, specify, and procure the hardware/software used in the system.

CAP question

R= r L/A

What is the resistance of 1000 ft of copper wire (specific resistance = 10.37) given a cross-sectional area of 10370 circular mil (cmil) and a wire temperature of 20°C?

A. 1 Ω

B. 2 Ω

C. 10 Ω

D. 100 Ω

CAP answer

The resistance of a length (L) of a conductor can be determined using the specific resistance and the cross-sectional area (A) in cmil by using the equation R = rL/A. The correct answer is **A, 1 Ω**.

Reference: Hughes, *Programmable Controllers*, ISA Press 2001.

Cmil is worth reviewing. Recall electrons flow through large-diameter wires easier than small-diameter wires, due to the greater cross-sectional area they have to move.

Rather than measure small wire sizes in inches, the unit of "mil" (1/1000 of an inch) is common. The cross-sectional area of a wire can be in terms of square units (square inches or square mils), circular mils, or gauge scale.

Calculating square-unit wire area for a circular wire involves the circle area formula:

$A=\pi r^2$ (Square units)

Calculating circular-mil wire area for a circular wire is much simpler because the unit of circular mil exists just for this purpose, to eliminate the pi and the d/2 (radius) factors in the formula.

$A=d^2$ (Circular units)

There are π (3.1416) square mils for every 4 circular mils.

1 October 2005

ISA Certified Control Systems Technician (CCST) Program

This question is from the Level III study guide, Domain 2, Loop checking.

Level III represents a professional who has a 13-year total of education, training, and/or experience.

CCST question

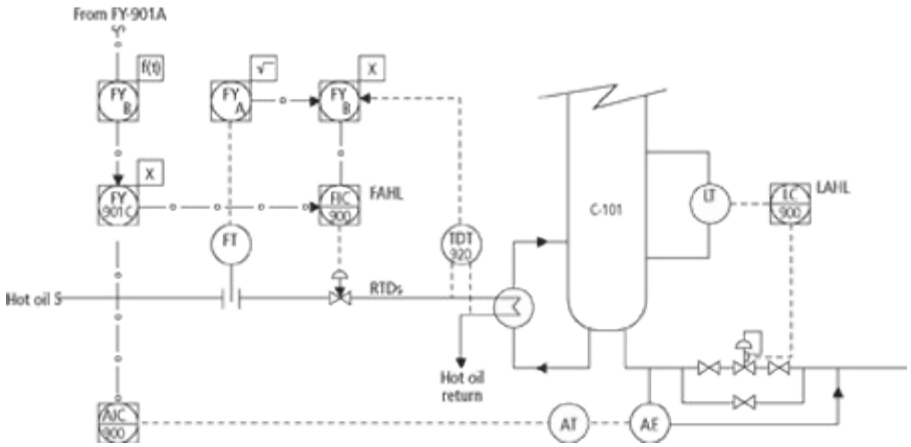
Refer to the figure. A loss of power to FIC 900 will result in the:

A. Level in C-101 to increase.

B. Level in C-101 to decrease.

C. Output of TDT 920 to increase.

D. Output of TDT 920 to decrease.



CCST answer

A loss of power to FIC 900, which is a flow controlling mechanism and opens and closes the valve, causes the valve to automatically close—Fail-Closed (FC). Symbols for failure modes are on the line between the valve symbol and the actuator symbol. The arrow pointing down, as it does here, indicates FC.

If the arrow had pointed up, that would indicate Fail-Open.

Therefore, with the valve closed, hot oil quits flowing through the line. The purpose of this flow is to transfer heat from the oil to the contents of the process vessel C-101.

TDT 920 measures the difference in the temperature of the hot oil flow before and after it contacts and transfers heat to the contents of C-101. Since there is no flow, the difference in temperatures of the two pipe sections move toward the same (ambient) temperature. Thus the difference in temperatures, and therefore TDT 920 output, decreases.

The correct answer is **D**.

1 October 2005

ISA Certified Automation Professional (CAP) Program

These questions are from the CAP study guide, Performance Domain III, System Design.

Definition: Design, specify, and procure the hardware/software used in the system.

CAP questions

1. An application includes a variable frequency drive (VFD) that implements a V/Hz method of torque control while maintaining a fixed V/Hz ratio. The motor has a nameplate rating of 460 volts and 60Hz. If the drive is operating at 30Hz, what percent of rated torque will the motor MOST likely develop?

A. 25%

B. 50%

C. 100%

D. 200%

2. To minimize electrical interference when AC power and DC signal wiring meet in a control panel, it is BEST to:

A. Use a different size wire.

B. Cross the wires at 90 degrees.

C. Run the wires parallel to each other.

D. Twist the AC wires around the DC wires.

CAP answers

1. The correct answer is **C**, 100%. The volts per hertz ratio (V/Hz) controls the torque of a motor. As long as this ratio is in proportion with the nameplate rating, the motor will develop rated torque. The only answer that represents rated torque is 100% rated torque. Reference: Polka, *Motors and Drives — A Practical Technology Guide*. ISA Press, 2003.

2. B, crossing the wires at 90 degrees is the correct answer. This minimizes the exposure of the wires to each other, and therefore minimizes the effects of interface. Using a different size wire will have no effect on the electrical interference. Running the wires

parallel to each other will increase the effects of interference. Twisting the AC wires around the DC wires will also increase the effects of interference. Reference: Hughes, *Programmable Controllers*. ISA Press, 2001.

CCST questions

1. All of the following allow specific information to be extracted from an electronic database EXCEPT:

A. Filter

B. Report

C. Template

D. Query

2. The continuous sequence of steps or "operating cycle" performed by a PLC processor is known as:

A. Updating

B. Scanning

C. Polling

D. Rectifying

CCST answers

These two questions test the CCST test taker's knowledge of nomenclature.

A *filter* is a program that accepts a certain type of data as input, transforms it in some manner, and then outputs the transformed data. For example, a program that sorts names is a filter because it accepts the names in unsorted order, sorts them, and then outputs the sorted names. Utilities that allow you to import or export data are also sometimes called filters.

In this context, a *report* is a formatted and organized presentation of data. Most database-management systems include a report writer that enables one to design and generate reports.

A *template* is something that establishes or serves as a pattern for reference.

A *query* is a request for information from a database. There are several different ways to pose a query.

Scanning is the process by which a computer collects data from process sensors for use in calculations. It is the sequential interrogation of devices or lists of information under the PLCs control.

While the PLC may in fact *update* values, *poll* devices, and *rectify* aspects of the process in response to data it encounters during its routine, the best term for this continuous *sequence* is *scanning*.

The correct answers are C, template, and B, scanning.

Definition: Design, specify, and procure the hardware/software used in the system.

CAP questions

1. How is PID defined in the phrase "PID control loop?"

A. Proportional Instrument Device

B. Piping Instrumentation Diagram

C. Percentage Integration Delta

D. Proportional Integral Derivative

2. P&ID is an acronym for which of the following?

A. Process and Instruction Document

B. Process and Instrument Diagram

C. Piping and Instrument Diagram

D. Piping and Installation Drawing

CAP answers

1. D, Proportional Integral Derivative

The PID term as described refers to the loop control technique used in controllers. Proportional Integral Derivative is the correct term as referenced. A PID Control Loop includes proportional, integral, and derivative functions to control the process.

Reference: Erickson and Hedrick, *Plantwide Process Control*, Wiley, 1999.

2. C: P&ID is a widely used acronym for Piping and Instrument Diagram.

Level III represents a professional who has a 13-year total of education, training, and/or experience.

CCST questions

1. Design Standards are defined as standards which:

A. Are least able to be enforced by regulatory bodies.

B. Are imposed by agencies such as OSHA, etc.

C. Have been agreed upon by various regulating bodies and refer to all aspects of defining quality and methods of fabrication or installation of materials and equipment.

D. Are developed by design companies.

2. Mandatory Standards are defined as standards:

A. Grouped within codes, supplemented by rules and regulations.

B. And conventions set by trade groups.

C. Which refer to content and presentation of documents.

D. None of the above

3. Data sheets:

A. Convey all useful information about an instrument.

B. Constitute a short form of specifications for an individual instrument or instrument type.

C. Contain necessary process data, physical maintenance requirements, environmental conditions, electrical requirements, and expected operational behavior for the instrument.

D. All of the above

CCST answers

The correct answers are: **1. C**, **2. A**, and **3. D**.

CAP question

The graphic display for a distillation column in a refining train would BEST fit in the HMI Display hierarchy at which level?

A. Plant

B. Area

C. Group

D. Loop

CAP answer

The best answer is **C**, Group.

A typical hierarchy is composed of displays at four levels:

Plant level – Provides information regarding the entire plant.

Area level – Provides information on portions of the plant equipment that have a working relationship.

Group level – Provides information for the control loops and data points relating to a single process unit within a plant area.

Loop level – Provides information that deals with individual control loops, sequences, or data points.

A distillation column refers to one piece of equipment in the distillation process that will contain several loops. Therefore, it falls into the group-level category.

1 January 2006

CAP, CCST, and CIMM Refreshers

ISA certification provides an objective, third-party assessment and confirmation of skills, and gives manufacturing and factory staff the opportunity to differentiate themselves from their peers and gain recognition. By encouraging certification of employees, employers also differentiate themselves by setting a high and consistent standard for job knowledge and performance resulting in a safer and more productive work environment.

In this column, *InTech* covers three areas of certification:

- **CAP – Certified Automation Professionals** are responsible for the direction, design, and deployment of systems and equipment for manufacturing and control systems.
- **CCST – Certified Control System Technicians** calibrate, document, troubleshoot, and repair/replace instrumentation for systems that measure and control level, temperature, pressure, flow, and other process variables.
- **CIMM – Certified Industrial Maintenance Mechanics** are responsible for preventive, predictive, and corrective maintenance. They are multi-skilled individuals whose expertise is primarily mechanical in nature as opposed to instrumentation or electrical.

ISA Certified Automation Professional (CAP) program

This question is from the CAP study guide, Performance Domain III, System Design.

Definition: Design, specify, and procure the hardware/software used in the system.

CAP question: To measure the level of a liquid with a dielectric constant greater than 2 in a horizontal vessel with changing composition and temperature, the device with the best reproducibility is a:

A. Radar level detector

B. Ultrasonic level detector

C. Capacitance level detector

D. Differential pressure transmitter

CAP answer: The fact that the tank is horizontal implies that the angle of the beam will be narrow at the surface and the vessel walls will not interfere. Radar can detect the smallest change in surface level. It does not depend upon the temperature or composition

of the fluid as long as the beam is narrow and the dielectric constant (the ratio of the capacitance of a material to the capacitance of air) of the fluid is greater than two.

Ultrasonic level detectors react to changes in the speed of sound with temperature and are not quite as sensitive as radar.

Capacitance level detectors react to changes in the dielectric constant with composition and are not quite as sensitive as radar.

Changes in liquid density with composition and temperature affect differential pressure transmitters and are not as sensitive as radar.

*Reference***:** Blevins, et al. *Advanced Control Unleashed: Plant Performance Management for Optimum Benefit*, ISA Press, 2003.

The best answer is **A**, radar level detector.

For information about the CAP program, go to www.isa.org/CAP.

ISA Certified Industrial Maintenance Mechanic (CIMM) program

CIMMs have a minimum of five years of relevant work experience in the maintenance mechanic field or three years experience and a two-year associate degree in maintenance or a related field.

CIMM questions:

1. The best way to assure that a full-face respirator has a positive seal is to place the palms of your hands:

A. Over the discharge and inhale

B. Over the inlet and inhale

C. On the front and press toward your face

D. On the bottom and push up

2. According to industry standards, at which minimum working height would an individual be required to wear a safety harness?

A. 3 feet

B. 4 feet

C. 5 feet

D. 6 feet

CIMM answers:

1. The best answer is **B**, over the inlet and inhale.

Before using a full-face respirator, an individual must perform a quantitative test to ensure a proper face seal and fit.

Reference: Tyco, Instructions for Scott V-2000 (Scott-O-Vista)

2. D, six feet or more, is the correct answer.

One's harness will not successfully deploy if one is working five feet or less off the ground and falls.

Reference: NCCER (National Center for Construction Education and Research), Core Curriculum: Trainee Guide, Prentice Hall, 2000.

For information about the CIMM program, go to www.isa.org/CIMM.

ISA Certified Control Systems Technician (CCST) program

These questions are from the Level III study guide, Domain 2, Loop checking. (Level III represents a professional who has a 13-year total of education, training, and/or experience.)

CCST questions:

1. Using ISA symbols on a P&ID (piping and instrument diagram), a solid line with three marks crossed by another hash mark represents:

A. Electric binary signal

B. Pneumatic binary signal

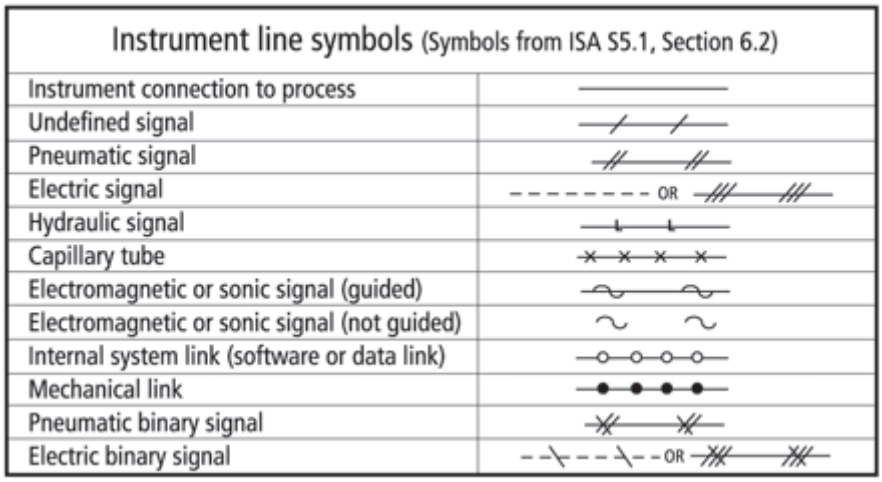
C. Undefined signal

D. Capillary tubing

CCST answer: *ISA-5.1-1984 - (R1992) Instrumentation Symbols and Identification* provides sufficient information to enable anyone reviewing any document depicting process measurement and control (who has a reasonable amount of process knowledge) to understand the means of measurement and control of the process.

The symbolism and identification methods provided in this standard are applicable to all classes of process measurement and control instrumentation. They can describe discrete instruments and their functions. They can also describe the analogous functions of systems that are variously termed shared display, shared control, distributed control, and computer control.

The best answer is **A**.



2. What part of a test procedure should be performed FIRST?

A. Operations test

B. Visual inspection

C. Output wiring check

D. Continuity test

CCST answer:

Looking at a mechanical and electrical situation globally to spot obvious and safety threats first is always best. Then move to more local and specific unit areas of the task. The best answer is **B**, visual inspection.

ISA Certified Control Systems Technician (CCST) program

These questions come from the Level I study guide, Domain 2, Loop checking. Level I (Level I represents a professional who has a five-year total of education, training, and/or experience.)

CCST questions

1. In most process-control systems, the final control element is a:
A. Transmitter
B. Sensor
C. Controller
D. Valve

2. A control valve with a direct acting actuator and direct trim will:
A. Fail closed
B. Maintain position
C. Fail open
D. Not fail

CCST answers: 1. The best answer is **D**, valve. A final control element is a device that directly controls the value of the manipulated variable of a control loop. Often the final control element is a control valve (ANSI/ISA-5.1-1984, revised 1992). It can also be the component of a control system, such as a control valve, that directly regulates the flow of energy or material to or from the process (ANSI/ISA-77.44.01-2000). Other examples of final control elements are solenoids and servomotors.

2. The best answer is **C**, fail open. A direct acting valve travels to the closed position when the signal increases. When there is a failure, there is no signal and power, and the direct acting valve fails open.

For information about the CCST program, go to www.isa.org/ccst.

ISA Certified Industrial Maintenance Mechanic (CIMM) program

CIMMs have a minimum of five years of relevant work experience in the maintenance mechanic field or three years' experience and a two-year associate degree in maintenance or a related field.

CIMM questions

1. When selecting the proper ladder for a task, what basic concept applies?
A. The distance from the ladder to your chin should be double the distance from the ladder to your knees.
B. The ladder chosen for the job should be wider than your shoulders.
C. The distance of the ladder from the building at the base should be 1/4 the working length of the ladder.
D. The ladder should be made of a different material than the material of the structure you are climbing.

2. Troubleshooting in a classified hazardous location where the work may be a source of ignition involves the use of a:

A. Hot work permit
B. Special nonflammable suit
C. Low-voltage multimeter
D. Second maintenance technician

CIMM answers: 1. The correct answer is C, the distance of the ladder from the building at the base should be 1/4 the working length of the ladder. While a four-to-one ratio is best practice, the ladder must also be at least 3 feet above the ladder contact point.

Reference: National Joint Steamfitter-Pipefitter Apprenticeship Committee, *Use and Care of Tools*, National Joint Steamfitter-Pipefitter Apprenticeship Committee, 1972.

2. The correct answer is A, hot work permit. Mechanics must take special care when utilizing tools or test equipment in a classified hazardous area. All tools must be properly rated for the environment in which they will be used to prevent the chance of causing a fire or explosion. A hot work permit confirms hazardous materials have been removed and ensures test equipment will not present a hazard.

Reference: Mostia, *Troubleshooting: A Technician's Guide*, ISA Press, 2000, www.isa.org/troubleshoot.

ISA Certified Automation Professional (CAP) program

These questions come from the CAP study guide, Performance Domain IV, Development.

Definition: Software development and coding.

CAP questions

1. When developing HMI screens, it is recommended that the quantity of items an operator has to remember is less than:
A. one
B. three
C. seven
D. 20

2. The average person cannot respond to signals more frequent than one every:
A. 30 milliseconds
B. 80 milliseconds
C. 300 milliseconds
D. 800 milliseconds

CAP answers: 1. The best answer is C, seven. Studies show people can remember approximately seven new things for about 20 seconds. This is called short-term memory.

After 20 seconds, people will have lost the information if they cannot quickly store it in long-term memory.

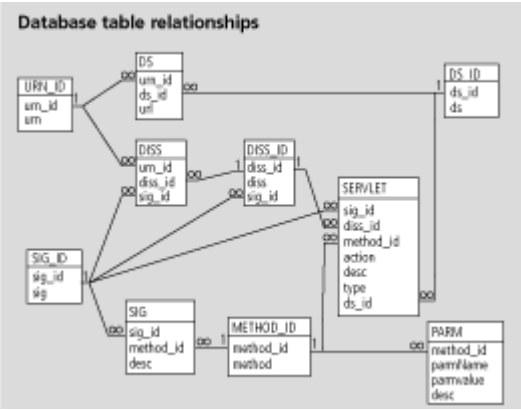
Reference: Weinschenk, et al. *GUI Design Essentials*, Wiley Computer Publishing, 1997.

2. The correct answer is **c**, 300 milliseconds. The average person cannot process signals more frequent than one every 300 milliseconds, or about three signals per second. Therefore, for reliable regulation of events that are faster than three signals per second, a person should not act as the controller.

ISA Certified Automation Professional (CAP) program

Certified Automation Professionals (CAPs) are responsible for the direction, design, and deployment of systems and equipment for manufacturing and control systems.

The following question comes from the CAP study guide, Performance Domain IV,



Development.

Definition: Software development and coding

CAP questions

1. A process data historian should be designed to:

A. Store data using an efficient data compression algorithm.
B. Store high-frequency process data online for a minimum of 10 years.
C. Store discrete values efficiently by combining bits into hexadecimal values.
D. Interpolate historical values to align time stamps.

2. Which database arranges data in a tabular format and links records by key values?

A. Hierarchical
B. Neutralized
C. Relational
D. Distributed

CAP answers

1. The correct answer is **A**, Store data using an efficient data compression algorithm.

Storing high frequency process data online for a minimum of 10 years (B) does not suggest any data-comparison techniques. Storing discrete values efficiently by combining bits into hexadecimal values (C) could result in lost time-stamp data.
Using interpolate historical values to align time stamps (D) would corrupt the actual data.

Reference: Fisher, Batch Control, ISA Press, 1996.

2. The correct answer is **C**, relational.

A relational database arranges information in a tabular format with each table, or relation, contributing data that describes a particular type of object. All connections between tables are based on shared key values.

In a hierarchical (A) database, information is maintained in a tree structure that emphasizes relationships between superior and subordinate items.

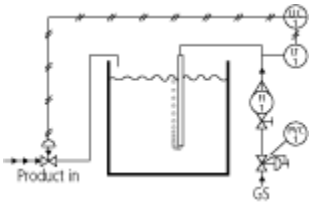
In a neutralized (B) database, predefined links connect many leaves in a tree. For information about the CAP program, go to http://www.isa.org/CAP.

ISA Certified Control Systems Technician (CCST) program

Certified Control System Technicians (CCSTs) calibrate, document, troubleshoot, and repair/replace instrumentation for systems that measure and control level, temperature, pressure, flow, and other process variables.

This question comes from the Level I study guide, Domain 2, Loop checking. Level I represents a professional who has a five-year total of education, training, and/or experience.

CCST question



1. In this "bubbler" application, FI-1:

A. Sets LIC-1 set point.
B. Controls purge gas flow.
C. Controls gas pressure.
D. Indicates purge gas flow.

CCST answer

November

The operation of a bubbler is similar to blowing air into a glass of water with a straw. The more water (higher water level) that is in the glass, the harder (more pressure) one needs to blow.

In this diagram, LIC is a level controller-indicating, LT is a level transmitter-blind, FI is a flow rate readout device-indicating, PCV is a pressure controller-self actuated control valve, and the GS indicates gas (purge) is flowing into the system. The correct answer is **D**.

ISA Certified Industrial Maintenance Mechanic (CIMM) program

Certified Industrial Maintenance Mechanics (CIMMs) are responsible for preventive, predictive, and corrective maintenance. They are multi-skilled individuals whose expertise is primarily mechanical in nature as opposed to instrumentation or electrical.

CIMMs have a minimum of five years of relevant work experience in the maintenance mechanic field or three years experience and a two-year associate degree in maintenance or a related field.

CIMM questions

1. You would use a thermocouple instead of a thermometer on boilers because thermocouples:

A. Can be used in a control circuit.
B. Have a slow response.
C. Are more expensive.
D. Can not handle higher heat.

2. Maintenance mechanics must work in noisy areas where hearing loss could occur. Which of the following statements about hearing loss is TRUE?

A. The shorter the duration of exposure, the greater the hearing loss.
B. The intensity of noise and duration of exposure determine the extent of hearing loss.
C. Hearing loss does not come from noise intensity.

CIMM answers

1. The correct answer is **A**.

Because thermocouples have electrical properties that change the temperature, control circuits can be employed to automatically make process adjustments or trip alarm points. Unlike a standard thermometer, a thermocouple uses an external display as a readout device. This allows a thermocouple to be centrally located in the process. Thermocouples have fast response times, are inexpensive, and typically handle higher ranges than thermometers.

Reference: Steingress, High Pressure Boilers, Homewood, IL: American Technical Publishing, 1994.

2. The correct answer is **B**.

When long-term exposure to intense noise occurs, the resulting hearing damage is referred to as an acute trauma or a chronic condition. Hearing loss is a serious health problem for the maintenance mechanic, and wearing hearing protection is the only safeguard available.

ISA Certified Automation Professional (CAP) program

Certified Automation Professionals (CAPs) are responsible for the direction, design, and deployment of systems and equipment for manufacturing and control systems.

The following question comes from the CAP study guide, Performance Domain IV, Development.

Definition: Software development and coding

CAP questions

Which PLC programming languages consists of contacts, timers, and counters?

A. Sequential Function Charts (SFC)

B. Ladder Diagram (LD)

C. Function Block Diagram (FBD)

D. Structured Text (ST)

In a typical PLC timer, what does the function of the preset value determine?

A. The resolution of the time base such as 1/10 sec or 1/100 sec

B. The length of time before the timer is energized or de-energized

C. The length of time the timer has been active

D. The on-delay or off-delay setting that determines the reset of the timer

CAP answers

1. The correct answer is **B**, Ladder Diagram (LD).

The LD Language consists of relay-type logic, timing, counting, and basic math operations. SFC and FBD are graphical languages in a high level structured language.

Reference: Lewis, *Programming Industrial Central Systems Using IEC 1131-3*, IEE Publishing, 1998.

2. The best answer is **B**, the length of time before the timer is energized or de-energized.

A timer pre-set is a programming value for the timing function. The resolution of the time base relates to accuracy. The length of time before the timer is energized or de-energized is a timer function.

Reference: Bailey and Wright, *Practical SCADA for Industry*, Newnes, 2003.

For information about the CAP program, go to www.isa.org/CAP.

ISA Certified Control Systems Technician (CCST) program

Certified Control System Technicians (CCSTs) calibrate, document, troubleshoot, and repair/replace instrumentation for systems that measure and control level, temperature, pressure, flow, and other process variables.

This question comes from the Level I study guide, Domain 2, Loop checking. Level I represents a professional who has a five-year total of education, training, and/or experience.

CCST questions

1. Which of the following types of actuators responds to a pneumatic signal?

A. Solenoid

B. Motor

C. Diaphragm

D. Electromagnetic valve

2. Changing the point of origin for input data and the output signal destination is easy with a Distributed Control System because many changes can be implemented by:

A. Relocating wire jumpers in a junction box.

B. Changing the pneumatic tubing connections.

C. Reassigning inputs/outputs by changing loop configuration.

D. Installing new transmitters and final control elements.

CCST answers

An actuator is a hydraulic, electric, or pneumatic device that moves an object or device.

Pneumatics (from the Greek pneumatikos, coming from the wind) is the use of pressurized gases to do work in science and technology.

The solenoid, the motor, and an electromagnetic valve all use electricity. A diaphragm expands and contracts depending on the amount of air pressure that exerts upon it.

The best answer to question 1 is **C**.

The best answer to question 2 is also **C**.

While one could accomplish changing data input and output by performing A, B, and D, the whole point of beauty in a distributed control system (DCS) is one doesn't have to go into the field to reconfigure the process.

Recall that a DCS is that class of instrumentation (input/output devices, control devices, and operator interface devices) that in addition to executing the stated control functions also permits the transmission of control, measurement, and operating information to and from one or many user-specifiable locations, all connected by a communication link. [ISA-5.3-1983]

For information about the CCST program, go to www.isa.org/ccst.

ISA Certified Industrial Maintenance Mechanic (CIMM) program

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CIMM questions

1. Which of the following is an example of a predictive maintenance task?

A. Changing oil on an hour meter set point.

B. Changing v-belts once a year on a fan drive.

C. Lubricating equipment on a set schedule.

D. Monitoring with vibration equipment.

2. Vibration analysis of equipment is used in predicting the possible failure of a component due to:

A. Observation of the equipment running.

B. The changing data collection points.

C. The improper equipment data collection.

D. The trending history of the component.

CIMM answers

1. The correct answer is **D**, monitoring with vibration equipment.

Vibration analysis is a form of predictive maintenance. The output curves of analysis assist in determining the condition of loose band wear on the equipment. Changing oil and v-belts and performing lubrication on equipment are preventive maintenance efforts.

Reference: Mobley, *An Introduction to Predictive Maintenance*. Woburn, MA, Butterworth-Heinemann, 2002.

2. The correct answer is **D**, The trending history of the component.

Vibration analysis can trend past history. It can be set to a trend mode to measure the rate of increase of a piece of equipment from a baseline reading to the present.

ISA Certified Automation Professional (CAP) program

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The following question comes from the CAP study guide, Performance Domain IV, Development.

Definition: Software development and coding.

CAP questions

1. Which of the following is NOT a strong source of electrical noise?

A. Variable speed drives

B. Switching power supplies

C. Electronic lighting ballasts

D. Regulated magnetic transformer power supply

2. Which of the following security technologies would BEST protect a manufacturing and control system from file transfer protocol access from outside the network?

A. Intrusion detection

B. Public key encryption

C. Password authentication

D. Firewall

CAP answers

1. The correct answer is **D**, Regulated magnetic transformer power supply.

A regulated magnetic-transformer power supply is the only choice that does not include power supply switching. Variable speed drives, switching power supplies, and electronic lighting ballasts all include switching power supplies, which we know produce noise in the system due to the high switching rate.

Reference: David Polka, *Motors & Drives: A Practical Technology Guide*, ISA Press, 2003.

2. The correct answer is **D**, Firewall.

Firewalls enforce access control policies using mechanisms that either block or permit certain types of traffic, thus regulating the flow of information. Public key encryption, password authentication, and intrusion detection are security techniques that block access to users.

Reference: ISA TR99.00.01 - 2004 - Security Technologies for Manufacturing and Control Systems, ISA, 2004.

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ISA Certified Control Systems Technician (CCST) program

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These questions come from the Level I study guide, Domain 3, Troubleshooting. Level I represents a professional who has a five-year total of education, training, and/or experience.

CCST question

Which type of flowmeter does a rotor mounted axially within a pipe between a set of bearings describe?

A. Magnetic

B. Turbine

C. Water meter

D. Venturi tube

CCST answer

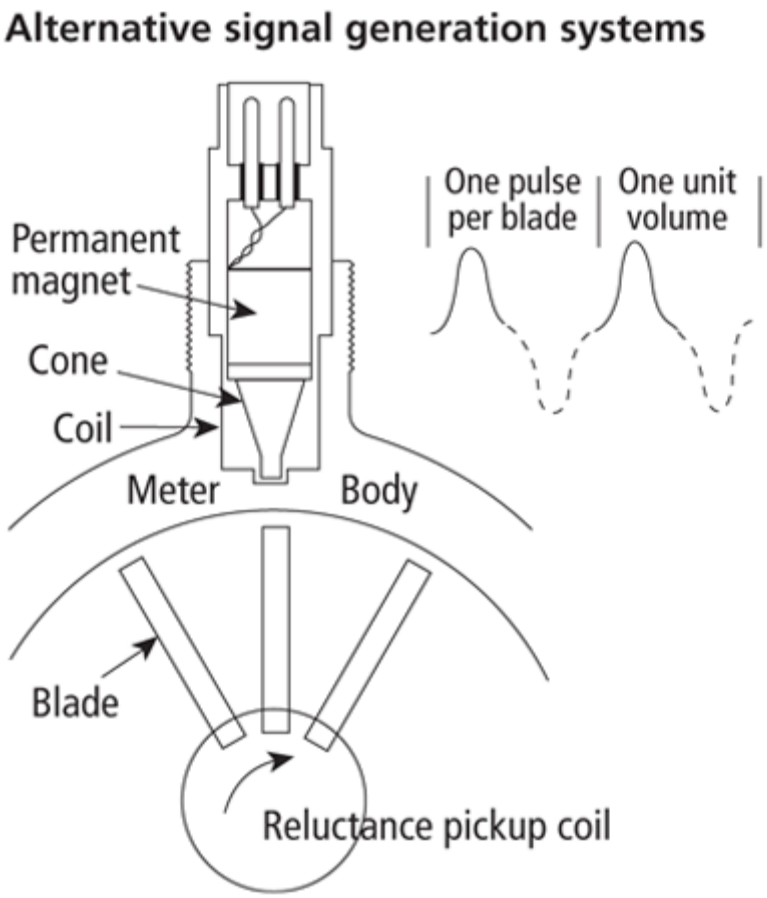
The correct answer is **B**.

A turbine meter has a multi-bladed rotor suspended in the fluid stream on a free-running bearing.

The axis of rotation of the rotor is perpendicular to the flow direction, and the rotor blades reach out nearly to the bore of meter. Think of the propeller of an airplane rotating and creating the flow of air that drives the plane. In the case of the turbine flowmeter, the flow of the fluid turns the propeller—turbine.

Therefore, the fluid impinging on the blades cases the rotor to revolve. The angular speed of the rotation is directly proportional to the volumetric flow rate.

An electromagnetic pickup coil monitors the speed of rotation as it sits on the outside of the meter housing. Because each blade of the turbine sweeps a discrete volume of fluid, each electrical impulse represents the same discrete volume of fluid.



For information about the CCST program, go to www.isa.org/ccst.

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These questions are from Performance Domain II: Preventive and Predictive Maintenance.

CIMM questions

1. The viscosity requirements of rolling-element bearings are dependent on the _____of the bearing.

A. Size, type, and operating temperature

B. Size, speed, and operating temperature

C. Style, size, and operating temperature

D. Style, type, and operating temperature

2. The alignment of suction and discharge piping and valves on an ANSI pump is critical to:

A. Ensure the correct coupling gap is achieved.

B. Reduce alignment issues with the coupling.

C. Reduce the stress on the coupling and shaft.

D. Reduce the amount of shim needed for alignment.

CIMM answers

1. The best answer to this question is **B**. Viscosity is the resistance of a liquid to flow. Thick liquids have high viscosity while thin liquids have low viscosity.

All bearings require lubrication to reduce friction. To determine the type of lubrication needed for a bearing, you need to know the size of the bearing and the speed and temperature of the equipment running.

2. The correct answer is **C**.

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The following questions come from the CAP study guide, Performance Domain IV, Development.

Definition: Software development and coding.

CAP questions

1. A binary number of 1101 has a hexadecimal equivalent of:

A. eight
B. 13
C. A
D. D

2. Which set of documents should be reviewed to gain a quick understanding of plant operations?

A. System architecture diagrams
B. Piping and Instrumentation diagrams
C. Plant layout drawing
D. Process flow diagrams

CAP answers

1. The correct answer is **D**. In hexadecimal, D is the equivalent of 1101 in binary. Eight in hexadecimal is the equivalent of 1000 in binary. Thirteen is not a hexadecimal format. A in hexadecimal is the equivalent of 1010 in binary.

Reference: Thomas A. Hughes, *Programmable Controllers*, ISA Press, 2001.

2. The best answer is **D**, process flow diagrams. They depict the flow of material through a plant, the major equipment, and their capacities. These diagrams make it very easy to visualize the plant as a whole.

Piping & Instrumentation diagrams come from PFDs and typically have too much detail to visualize the entire plant. Plant layout drawings provide a visual representation of the physical locations of buildings, major structures, and major equipment, much like a map. System architecture diagrams show the interconnection of all the major control system components. They do not provide much information on how the plant works.

Reference: Bela Liptak, editor, *Instrument Engineers' Handbook: Volume 3 (3rd Edition) Process Software and Digital Networks*. CRC Press and ISA Press, 2002.

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ISA Certified Control Systems Technician (CCST) program

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This question comes from the Level I study guide, Domain 3, Troubleshooting. Level I represents a professional who has a five-year total of education, training, and/or experience.

CCST question

Which type of flowmeter measures flow by measuring volume directly?

A. Coriolis
B. Magnetic
C. Positive displacement
D. Differential pressure

CCST answer

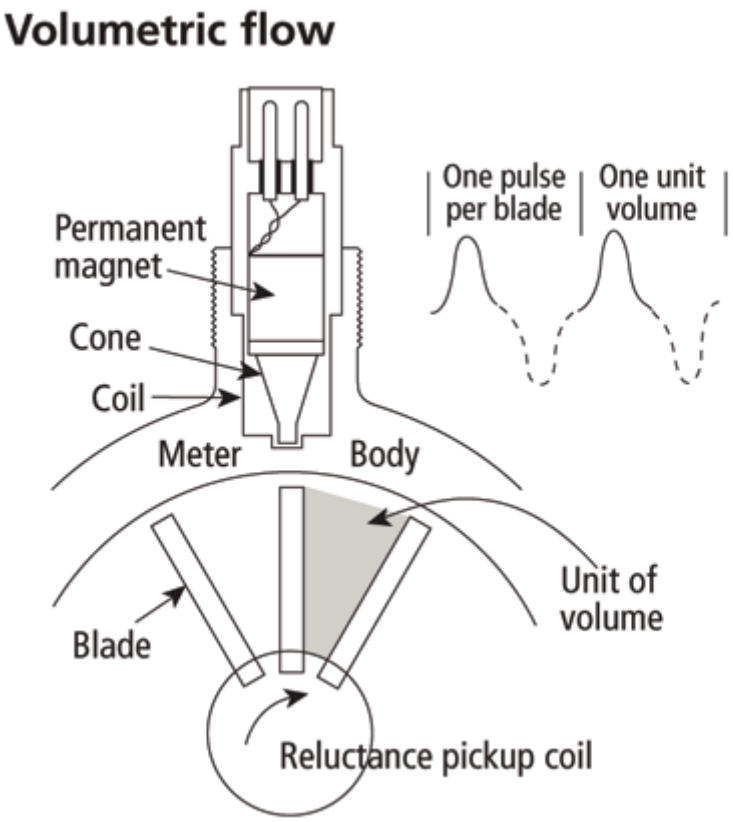
A *Coriolis*-type mass flowmeter measures the flow rate by determining the torque caused by the radial acceleration of the fluid. It does not measure the volume of the liquid passing through the tube; it measures the amount of mass flowing through the device.

A *magnetic* flowmeter is a device for measuring volumetric flow rate across a fluid stream by setting up a magnetic field perpendicular to the direction of flow. A flowing conductive fluid then generates a voltage proportional to fluid speed.

A *positive-displacement* flowmeter measures the flow by breaking it up into discrete elements or packages of fluid. Each package has a known volume. One determines the flow rate by counting how many discrete elements pass through the meter per unit time.

A *differential pressure* flowmeter (dP) uses a primary element to place a constriction in the flow stream, causing a drop in pressure. Then one uses the resulting pressure drop to calculate the flow rate. Flow rate is proportional to the square root of the difference in pressure one can calculate it using Bernoulli's equation.

Of these four methods only c—positive displacement—measures the volume directly.



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These questions are from Performance Domain II: Preventive and Predictive Maintenance.

CIMM questions

1. The main purpose of a water column on a steam boiler is to:

A. Be able to blow down the gauge glass.
B. Reduce turbulence in the gauge glass.

C. Provide a mounting location for high and low alarms.
D. Provide a mounting location for a gauge glass.

2. What is Net Positive Suction Head (NPSH)?

A. The suction pressure measured at the tank feeding the pump
B. The height of the liquid available to feed the pump
C. The amount of vacuum on the suction of the pump
D. Running suction pressure on the pump to prevent cavitation

CIMM answers

1. The best answer is **B**, reduce turbulence in the gauge glass.

The gauge glass on a boiler is a visual display of water level in the boiler drum. The graduated gauge mounts to a parallel column of boiler water, which is a much larger diameter than the gauge glass. As the water level in the boiler rises and falls, the water level in the water column and the gauging glass rise and fall together. Turbulence inside the boiler is less in the water column, so an accurate water level can be determined on the gauge glass.

2. The correct answer is **B**, the height of the liquid available to feed the pump.

The NPSH is the height of the liquid within a tank that is allowable for the pump to draw feed into the suction. If you do not have the NPSH required over the NPSH acquired, your pump will starve for fluid.

ISA Certified Automation Professional (CAP) program

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The following questions come from the CAP study guide, Performance Domain V, Deployment.

Field installation and checkout and startup of the systems

CAP questions
1. A diagram that shows all electrical devices and wiring details specific to a particular physical location is commonly referred to as a(n) _____ diagram.

A. Arrangement
B. Termination

C. Loop
D. Schematic

2. For a rapid and representative pH measurement at the outlet of an inline neutralizer, such as a static mixer, the electrode should be approximately _____ pipe diameters downstream of the outlet.

A. 0.2
B. 2
C. 20
D. 200

CAP answers
1. The correct answer is **B**, a termination diagram.

The termination diagram's purpose is to show all electrical devices and wiring details specific to a particular physical location.

Arrangement drawings show the dimensional mounting arrangement of all devices specific to a particular physical location and to prove installation specifics and Bill of Material (listing of parts of an item as shown in a drawing).

A loop diagram's purpose is to use the process function as the focal point and show all wiring and devices required to accommodate that function.

A scheme diagram shows electrical devices and wiring details.

2. The correct answer is **C**, 20 pipe diameters.

Equipment in a pipeline, such as static mixers, split up, or stratify streams, that need some distance to recombine to reduce the noise in the measurement.

Twenty pipe diameters will provide enough mixing in most cases for turbulent flow and liquid mixing.

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This question comes from the Level I study guide, Domain 3, Troubleshooting. Level I represents a professional who has a five-year total of education, training, and/or experience.

CCST question
What occurs if the temperature of the thermocouple measuring junction is lower than the reference junction?

A. There is no emf output.
B. The output voltage polarity is reversed.
C. The polarity stays the same, but voltage increases.
D. The emf remains the same when temperature changes.

CCST answer
A thermocouple works on a principal that Thomas Seebeck discovered in 1821. He found when any conductor—metal—experiences a difference in temperature, it would generate a voltage.

When it connects another conductor to the "hot" end of the first conductor, an additional conductor will then also experience the temperature gradient and develop a voltage of its own which will oppose the original.

The magnitude of the effect depends on the metal in use. Using a dissimilar metal to complete the circuit will have a different voltage generated, leaving a small difference voltage available for us to measure, which increases with temperature.

This difference can typically be between one to about 70 microvolts per degree Celsius for the modern range of available metal combinations.

Thermocouples measure the temperature difference between two points not absolute temperature. Whether one junction is hotter or colder than other doesn't matter, only the direction of the voltage flow changes. Polarity reverses (**B**) is the correct answer.

For information about the CCST program, go to www.isa.org/ccst.

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These questions are from Performance Domain I: Maintenance practices.

CIMM questions
1. What is the fraction 1/8 equal to when converted into a decimal?

A. .0625
B. .093
C. .125
D. .375

2. Using the Pythagorean theorem (A2+B2=C2), compute the length of the hypotenuse of the triangle at right.

A. 5”
B. 5.5”
C. 5.75”
D. 6”

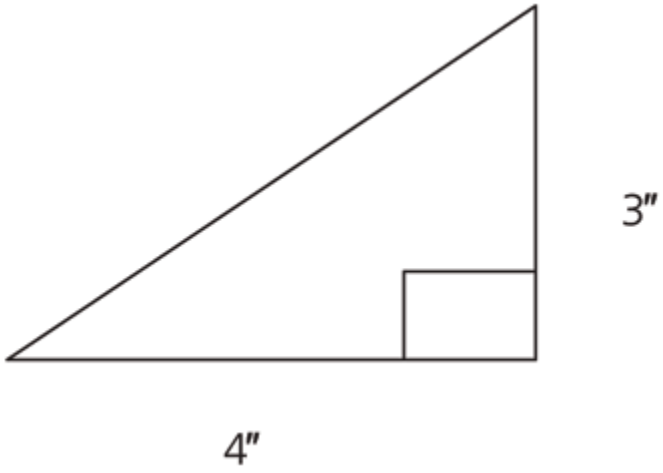
CIMM answers
1. **C, .125,** is the correct answer. One is divided by eight to convert 1/8 to a decimal.

Reference: NCCER, Core Curriculum: Trainee Guide. Upper Saddle River, NJ. Prentice-Hall, 2000.

2. **A, 5”** (five inches), is the correct answer.

Pythagorean is a common formula for determining length of pipe. To calculate the hypotenuse use

$$C^2 = A^2 + B^2 \quad \text{therefore} \quad C^2 = 3^2 + 4^2$$
$$\text{and} \quad C^2 = 9 + 16$$
$$\text{so} \quad C^2 = 25, \sqrt{C^2} = \sqrt{25}$$
$$C = 5$$



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These questions are from Performance Domain I: Maintenance practices

CIMM questions

1. To avoid a fire hazard, what must be kept away from grinding operations?
A. Passersby
B. Combustibles
C. Metal stock
D. Sparks

2. What is used to cut internal threads on a pump base?
A. A tap
B. A die nut
C. A collet
D. A reamer

CIMM answers

1. The best answer is B, combustibles.
Sparks sometimes occur during grinding operation and when grinding metals. Keeping combustibles away from this operation will help avoid fires.
Reference: Althouse, Modern Welding, Tinley Park, IL, Goodheart-Willcox, 1997

2. A, tap is the correct answer.
A tap has cutting flutes made of hardened metal. A precise tap hole is pre-drilled, and the tap is turned into the hole, cutting internal threads as it moves. Special cutting oil is applied to reduce friction, resulting in more accurate threads being cut. A mechanic will often use a tap wrench to turn the tap by hand, moving a little amount at a time, backing up slightly, and then advancing again into fresh metal.
Reference: National Joint Steamfitter-Pipefitter Apprenticeship Committee, *Use and Care of Tools*. Washington, DC, 1972

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This question comes from the Level I study guide, Domain 3, Troubleshooting. Level I represents a professional who has a five-year total of education, training, and/or experience.

CCST question
A ground loop on shielded cable can be avoided by:
A. Attaching a safety ground to the instrument casing
B. Grounding the shield at only one end of the circuit
C. Attaching a safety ground to the control panel
D. Grounding the shield at both ends of the cable

CCST answer
A ground loop refers to an unwanted electrical current in a conductor connecting two points that are supposed to be at the same potential (units are volts), but that are actually at different potentials (not grounded). Ground loops are detrimental to the intended operation of the electrical system.
Potential is the difference in voltage between two points. One of the points should be a ground with voltage equal to 0 (zero).
The best way to avert the ground loop in this question is to ground the shield at only one end of the circuit. The correct answer is B.
For information about the CCST program, go to www.isa.org/ccst

ISA Certified Automation Professional (CAP) program

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The following questions come from the CAP study guide, Performance Domain V, Deployment.

Field installation and checkout and startup of the systems

CAP questions

1. Considering a 60 Hz power source, at what speed does a four-pole synchronous motor run?

A. 1200 RPM

B. 1800 RPM

C. 2400 RPM

D. 3600 RPM

2. When a large automation project makes significant changes to plant operations, it is BEST to perform operator training:

A. Six months prior to the startup phase during control system development

B. After acceptance testing and prior to the startup phase

C. During the startup phase after water batching

D. After the startup phase is complete and the process has stabilized

CAP answers

1. The correct answer is B, 1800 revolutions per minute (RPM).

Synchronous speed in RPM = (two × frequency × 60) / (number of poles)

For a 60 Hz, four pole motor that would be = (two × 60 × 60) / four = 1800

Reference: Webb, Industrial Control Electronics, Merrill Publishing, 1990

2. The correct answer is B, after acceptance testing and prior to the startup phase.

The best time to train operators is after acceptance testing and prior to startup. Ideally, you want to train the operators after the bulk of the changes have happened, but before they will need to implement the knowledge. This reduces confusion and also reduces the chance that training will exit operator memory. Training needs to occur before the startup phase because the operator would have to know something about the process before they could start up the plant. However, if training occurs too early, the operator will forget what he or she learned before having a chance to apply the knowledge.

Documenting skills is value-add

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Field installation and checkout and startup of the systems

CAP questions

1. If an exothermic reactor can run-away in manual, then the temperature controller should be tuned with a(n):

A. Open loop short cut method
B. Closed loop
C. Lambda Tuning method
D. Reaction curve method

2. An increase in which setting can compensate for a thermowell lag time?

A. Rate time
B. Reset time
C. Gain
D. Filter time

CAP answers

1. The best answer is **B**, closed loop.

The controller must stay in automatic to prevent a runaway reaction. Closed-loop methods keep the controller in automatic. Normally, we maximize the gain to the point where either the error is small enough or the process starts to show a slight oscillation. It does not imply the ultimate oscillation method. The controller is in manual for *open loop*, *Lambda Tuning*, and *reaction curve* methods.

Reference: McMillan, *Good Tuning: A Pocket Guide*, ISA Press, 2000.

2. The correct answer is **A**, rate time.

Thermowell and temperature processes have multiple interacting thermal lags. Rate time cancels a secondary thermal lag that can improve control in a slow temperature process without A/D chatter. Reset time adds another lag that further increases the apparent dead time from multiple lags. A large thermowell lag reduces the allowable gain. A process variable filter adds another lag that further increases the apparent dead time from multiple lags.

Reference: McMillan, *Good Tuning: A Pocket Guide*, ISA Press, 2000.

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Source: ABB

These questions comes from the Level I study guide, Domain 3, Troubleshooting. Level I represents a professional who has a five-year total of education, training, and/or experience.

CCST questions

1. A low resistance reading from an ungrounded signal lead connected to a ground indicates a problem of:

A. A lead having an unwanted connection to ground
B. An open circuit from the lead to ground
C. An open circuit from the lead to the casing
D. None of the above

2. What kind of wires in a process control loop would most likely be shielded cables?

A. Power wires
B. Ground wires
C. Signal wires
D. Control wires

CCST answers

The correct answers are **A** for question number one and **C** for question number two.

Potential is the difference in voltage between two points. One of the points should be a ground with voltage equal to 0 (zero). The best way to avert the ground loop in this question is to ground the shield at only one end of the circuit. The correct answer is B.

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These questions are from Performance Domain I: Maintenance practices.

CIMM questions

1. Prior to operating a powered industrial lift truck in the plant, each operator must successfully complete:

A. An approved training program
B. The accompanying work order
C. All non-essential maintenance tasks first
D. A hot work permit

2. Selecting a ladder of sufficient length is important in order to avoid:

A. Door openings
B. Unnecessary climbing
C. Electric power lines
D. Stretching or reaching

CIMM answers

1. The correct answer is **A**, an approved training program.

OSHA requires all operators of powered industrial lift trucks undergo classroom and field-testing. Because of the hazards presented by the elevated loads while in motion, it is imperative operators fully understand how properly to handle equipment and loads. The operator must also demonstrate the ability to properly inspect and operate powered industrial lift trucks before receiving his or her certificate.

Reference: IBEW-NECA, OSHA Requirements for General Industry, IBEW-NECA, 2003.

2. D, stretching or reaching, is the correct answer.

A ladder will easily tip over if you reach outside the centerline.

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Field installation and checkout and startup of the systems

CAP questions

1. A Bill of Materials for purchase of cable on a project is created by completing a material:

A. Procurement
B. Inspection
C. Take off
D. Requisition

2. If the distance in a sheet line from the thickness gauge to the manipulated diebolt actuators is 200 meters and the speed is 4 meters per second, then the dead time from the transportation delay for thickness control is _____ seconds.

A. 200
B. 50
C. 5
D. 0.02

CAP answers

1. C, Take off, is the correct answer. A material takeoff is the process of analyzing the drawings and determining all the materials required to accomplish the design. We then use the material takeoff to create a Bill of Materials. Inspection does not aid in creating a Bill of Material. Procurement and requisition are activities that occur after the Bill of Materials is complete. *Reference:* Whitt*, Successful Instrument and Control System Design, ISA Press, 2004.*



2. B, 50 is the best answer. The time it takes the sheet to go from the diebolt actuators, which are the manipulated variable, to the thickness gauge that transverses the sheet to provide the controlled variable (sheet thickness profile across the sheet), is the transportation delay. 50 seconds: Distance / velocity (200 meters / 4 meters per second) is the transportation delay. *Reference:* McMillan*, Good Tuning: A Pocket Guide, ISA Press, 2000* and *Blevins, et al., Advanced Control Unleashed: Plant Performance Management for Optimum Benefit, ISA Press, 2003.*

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Certified Control System Technicians (CCSTs) calibrate, document, troubleshoot, and repair/replace instrumentation for systems that measure and control level, temperature, pressure, flow, and other process variables.

These questions comes from the Level I study guide, Domain 3, Troubleshooting. Level I represents a professional who has a five-year total of education, training, and/or experience.

CCST questions

1. A device used to change an analog signal to a digital signal is most often called a:

A. Transducer
B. Signal conditioner
C. Transmitter
D. Converter

2. A bread boarding area is:

A. A data routing circuit
B. Used to design or test electronic circuits
C. A code-converting circuit
D. A temporary binary storage area

CCST answers

The answer to the first question is **D**, a converter. A bread boarding area is a place to design or test electronic circuits. The answer to number two is **B**.

For information about the CCST program, go to www.isa.org/ccst.

ISA Certified Industrial Maintenance Mechanic (CIMM) program

Certified Industrial Maintenance Mechanics (CIMMs) are responsible for preventive, predictive, and corrective maintenance. They are multi-skilled individuals whose expertise is primarily mechanical in nature as opposed to instrumentation or electrical.

CIMMs have a minimum of five years of relevant work experience in the maintenance mechanic field or three years experience and a two-year associate degree in maintenance or a related field.

These questions are from Performance Domain I: Maintenance practices.

CIMM questions

1. Which of the following is an example of a maintenance activity that requires a lookout/tagout for the maintenance task(s)?

A. Removing or bypassing a guard to perform a maintenance task on a piece of equipment
B. Centering and adjusting a belt on a conveyor system
C. Identifying problems and verifying corrections during troubleshooting activities
D. Performing minor tool changes and adjustments on a milling machine

2. In OSHA regulations, the term “point-of-operation protection device” refers to what feature in maintenance?

A. Lockout-tagout
B. Machine guarding
C. Personal protective equipment
D. Pre-operation inspection

CIMM answers

1. The correct answer is A, removing or bypassing a guard to perform a maintenance task on a piece of equipment. If the guard is removed or any moving materials are exposed, the equipment must go through a proper lock, tag, and try procedure.

Reference: U.S. Department of Labor, OSHA 3.20. U.S. Department of Labor, 1997.

2. The answer is **B**, machine guarding. The machine guard is the first protective device on most pieces of equipment. Machine guards are in place to protect the user and mount directly to the equipment.

"Lockout/tagout" and "PPE" are not attached-to-the-equipment related. A "pre-operation inspection" would inspect if the guard was in place and serviceable.